KEN4256 — Research Project Report | March 2026

Global Ownership Structures Behind Modern Conflicts: A Multi-War Knowledge Graph Analysis

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March 2026

Abstract

The corporate ownership structures behind military production remain a structural blind spot in conflict analysis: while the global arms trade is well-documented at the state-to-state level, the transnational hierarchies of parent companies and holding groups that ultimately control manufacturers are rarely mapped, and never systematically compared across multiple wars. This paper addresses that gap by constructing and mining a multi-conflict ownership knowledge graph that links military assets to their manufacturers and ultimate controlling entities across nine contemporary armed conflicts (Ukraine, Gaza, Syria, Iraq, Afghanistan, Lebanon 2006, Kosovo 1999, Nagorno-Karabakh, and the Iran–Iraq War), using Wikidata as the sole linked-data source. A modular SPARQL extraction and ownership traversal pipeline produced a dataset of 835 conflict–asset–manufacturer triples linked to 145 ultimate controlling entities. Network and concentration analysis on the resulting ownership graph reveals a pronounced asymmetry: the United States and Russia emerge as systemic players with a presence across the majority of sampled conflicts, ownership concentration varies widely across wars, and community detection recovers clusters that closely mirror known geopolitical alignments. These results demonstrate that a small set of corporate and state actors exerts systemic cross-conflict industrial influence that is structurally invisible to traditional trade-based metrics, and that knowledge graphs offer a principled, reproducible framework for making that influence legible.

1. Introduction

Conflict analysis has long focused on the political, military, and humanitarian dimensions of armed conflict. A structurally distinct dimension, the corporate ownership of the industrial base that produces military equipment, has received far less systematic attention. While the global arms trade is well-documented at the state-to-state level through databases such as SIPRI, the vertical hierarchies linking individual manufacturers to their parent companies and ultimate controlling entities remain largely opaque. This matters because the global defense industry is highly concentrated: a relatively small number of corporate groups control the majority of production, and the same entities may exert influence across multiple, geographically and politically unrelated conflicts simultaneously. Without a structured mapping of these hierarchies, analyses of warfare overlook the economic incentives and industrial overlaps that sustain military supply over time.

This project addresses that gap through a knowledge-graph-based approach. By integrating Wikidata's linked representations of conflicts, military assets, manufacturers, and ownership relations, the study constructs a cross-conflict ownership network and subjects it to graph-theoretic and economic concentration analysis.

Two research questions guide the project:

- **RQ1:** Who are the ultimate controlling entities behind the companies manufacturing military assets used in modern warfare?
- **RQ2:** How concentrated is that ownership, and to what extent do the same corporate groups exhibit a systemic presence across geographically and politically distinct conflicts?

The specific objectives are to: (i) build a structured ownership dataset linking assets, manufacturers, and controllers across nine conflicts; (ii) produce comparative ownership profiles per conflict; (iii) analyse the resulting network through centrality measures, community detection, and economic concentration metrics (HHI, top-k ownership share); and (iv) generate interpretable visual outputs to support systematic comparison. Section 2 surveys related work; Section 3 describes the methodology; Section 4 presents results; Section 5 discusses their significance; Section 6 concludes.

2. Related Work

Research on the defense sector and armed conflict draws on three largely disconnected bodies of literature. The Stockholm International Peace Research Institute (SIPRI) provides the standard reference for firm-level arms revenues and bilateral transfer flows [1, 2, 3]. SIPRI's data are authoritative at the transaction level but do not represent the vertical ownership hierarchies linking subsidiaries to ultimate controlling entities, nor do they formally integrate with conflict participation data.

On the conflict side, the Uppsala Conflict Data Program (UCDP) and the Armed Conflict Location and Event Data Project (ACLED) offer high-resolution records of violence patterns and participant dynamics. Like SIPRI, these datasets operate in isolation from industrial ownership information, making cross-domain inference structurally impossible without additional integration effort.

Within network science, Lebacher et al. [4] propose a

dynamic separable network model for global weapons transfers incorporating actor heterogeneity, while Jang and Yang [5] analyse the structural stability of the global arms trade network. Both studies advance the graph-analytic treatment of the arms trade but stop at the manufacturer level, leaving higher-order ownership hierarchies unexplored.

Knowledge graph approaches have been applied to corporate ownership in financial contexts (e.g., beneficial ownership mapping in anti-money-laundering research), but their application to the defense sector and conflict networks is novel. This project bridges these silos by using Wikidata as a unified linked-data substrate, a design choice that enables seamless traversal from conflict entity to asset to manufacturer to controlling group within a single RDF graph, without requiring cross-database reconciliation.

3. Methodology

3.1 Data Source and Technology Stack

All data were extracted from Wikidata via the Wikidata Query Service (WDQS) using SPARQL. Wikidata was chosen as the sole data source because it encodes, within a single interoperable RDF graph, the full chain of entities required by this study: conflicts , military assets, manufacturers, and corporate ownership relations. This eliminates the need for cross-source entity reconciliation, which would otherwise require manual disambiguation at the manufacturer level.

The processing stack comprised Python with `pandas` for tabular manipulation, `requests` for batched SPARQL calls to the WDQS endpoint, `networkx` for graph construction and metric computation, `python-louvain` for community detection, and `seaborn`/`matplotlib` for visualisation. All intermediate results were serialised to CSV and the final knowledge graph to Turtle (RDF).

3.2 Conflict and Asset Extraction

Nine conflicts were selected on the basis of Wikidata coverage and contemporary relevance: the Russo-Ukrainian War (Q110999040), the Gaza War 2023 (Q122962941), the Syrian Civil War (Q178810), the Iraq War (Q545449), the War in Afghanistan (Q182865), the 2006 Lebanon War (Q49104), the Kosovo War (Q190029), the 2020 Nagorno-Karabakh War (Q99659280), and the Iran–Iraq War (Q82664). Yemen and Libya were assessed as candidates but excluded due to insufficient P710→P137 linkage in Wikidata.

Figure 1 illustrates the five-layer graph structure constructed by the pipeline, from conflict entities down to ultimate ownership.

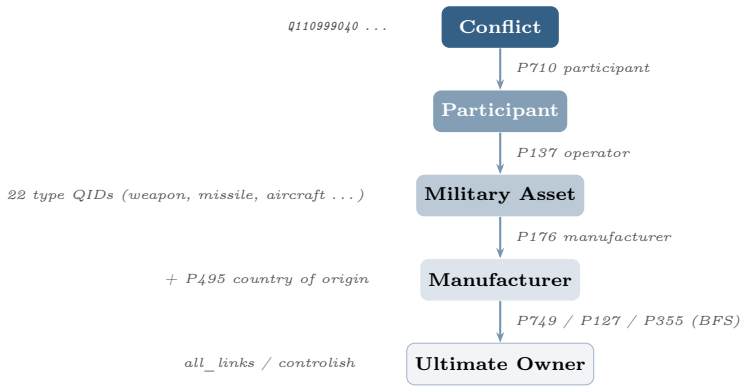


Figure 1: Five-layer knowledge graph structure. Each arrow denotes the Wikidata predicate used to traverse from one layer to the next. BFS ownership expansion (bottom arrow) iterates upward until a semantic terminal (sovereign state, natural person) or `MAX_DEPTH` = 5 is reached.

For each of these nine conflicts, a SPARQL query retrieved participants via P710 and linked them to military assets via the P137 (*operator*) predicate. Assets were filtered to a controlled type vocabulary of 22 Wikidata class QIDs covering weapons, artillery, missiles, aircraft, combat vehicles, and naval vessels (e.g. `Q15142894` military aircraft, `Q913` missile, `Q12876` tank). Entities matching none of these types were discarded. To avoid WDQS timeout errors, queries were issued per conflict and results cached locally before merging.

For each retained asset, P176 (manufacturer) was extracted, along with P495 (country of origin) for the manufacturer entity itself. After deduplication across the nine per-conflict files, the merged dataset comprised 835 unique conflict–asset–manufacturer triples (reduced from 1,013 raw rows). Manufacturer coverage was checked; the share of assets with at least one P176 link ranged from 75.3% (Gaza) to 94.9% (Iraq), reflecting variation in Wikidata curation depth across conflicts.

3.3 Ownership Chain Reconstruction

Iterative BFS expansion. Starting from the 194 unique manufacturers identified in M2, ownership edges were retrieved from Wikidata using an iterative frontier-based strategy. At each iteration, the current frontier (initially the 194 seed manufacturers) was queried in batches of 80 entities. For each entity, all outgoing ownership predicates were fetched: P749 (*parent organisation*, 158 edges), P127 (*owned by*, 259 edges), P355 (*subsidiary of owner*, 119 edges), and a fallback via P159 (*headquarters location*) when no direct ownership link was available (117 edges). P361 (*part of*) was excluded because it connects entities through non-ownership relations (e.g. military units to armed forces, companies to industrial sectors) and produced high noise in preliminary runs (164 edges removed).

Newly discovered owner entities not yet in the graph were added to the next frontier. Sovereign states, coun-

tries, and natural persons (identified by their P31 *instance of* values, e.g. Q3624078 *sovereign state*, Q5 *human*) were flagged as semantic terminals and removed from the frontier, since querying for their parent organisations yields no meaningful corporate hierarchy. The traversal ran for five iterations before hitting the `MAX_DEPTH` cap, discovering 145, 86, 64, 37, and 13 new owners at successive depths for a total of 817 edges across 525 entities.

Ultimate owner resolution. BFS was then run on the resulting edge graph to identify, for each seed manufacturer, the set of terminal nodes reachable by upward traversal. Two resolution modes were retained in parallel: *all_links*, which follows all fetched predicates, and *controlish*, which restricts traversal to predicates with a stronger control semantics (P749, P355, and directional P127 edges), excluding passive financial links. Terminal nodes mapped to semantically generic classes, including *institutional investor*, *investor*, *bank* and historically implausible entities, were flagged as noise and removed before analysis. The clean ownership table was saved as the primary input.

3.4 Network Construction and Analysis

Graph construction. A bipartite Owner–Conflict graph B was constructed in `networkx` using the *controlish* ownership table. Nodes are partitioned into conflict nodes (9, corresponding to the nine conflicts with sufficient asset coverage) and owner nodes (74). A weighted edge connects owner o to conflict c with weight equal to the number of distinct manufacturers controlled by o that appear in c. The full directed ownership graph G (431 nodes, 520 edges) was retained separately for centrality computation.

Centrality. Three centrality measures were computed on G: in-degree centrality (number of subsidiaries pointing to a node), betweenness centrality (normalised fraction of shortest paths passing through a node), and PageRank with damping factor $\alpha = 0.85$ and edge weights proportional to the number of manufacturers controlled. PageRank was preferred over simple degree for the summary analysis because it propagates influence recursively through the ownership graph, giving higher scores to entities that control other well-connected entities rather than simply many leaf manufacturers.

Community detection. The Louvain algorithm was applied to the projected owner–owner network (74 nodes, 1,932 edges), where two owners are connected if they share at least one conflict. Louvain optimises modularity, grouping owners that co-occur in conflicts more densely than expected under a null model.

Concentration metrics. Ownership concentration within each conflict was quantified using the Herfindahl–Hirschman Index:

$$HHI = \sum_{i=1}^{n} s_i^2$$

where s_i is the share of manufacturers within conflict c controlled by owner i. HHI ranges from $1/n$ (perfectly equal distribution) to 1 (monopoly). It was computed under both modes: *all_links* aggregated by country of the ultimate owner (measuring state-level concentration) and *controlish* at the individual conglomerate level. The top-1 and top-3 ownership shares were computed alongside HHI as complementary measures.

Conflict similarity. To measure how much two conflicts share the same ownership base, Jaccard similarity was computed for each conflict pair using the binary owner–conflict matrix (*controlish* mode):

$$J(C_i, C_j) = \frac{|\text{owners}(C_i) \cap \text{owners}(C_j)|}{|\text{owners}(C_i) \cup \text{owners}(C_j)|}$$

$J = 1$ indicates identical ownership sets; $J = 0$ indicates no shared owners. The resulting 9×9 matrix was hierarchically clustered using Ward linkage to reveal the underlying ownership-based grouping of conflicts.

3.5 Risk Mitigation

Three risks identified in the project proposal were addressed as follows. *Incomplete or misleading asset coverage*: mitigated by adopting a broad type vocabulary (22 QIDs) and retaining both strict and expanded asset definitions for sensitivity comparison. *Incomplete ownership chains*: mitigated by combining five ownership predicates and applying the P159 headquarters fallback, guaranteeing at least a jurisdiction-level proxy for every manufacturer. *WDQS timeout and rate limits*: mitigated by per-conflict modular extraction, batched queries (batch size 80), a 10-second delay between conflict downloads, and full local CSV caching of all intermediate results.

4. Results

4.1 Dataset Overview

The extraction pipeline produced 835 unique conflict–asset–manufacturer triples (post-deduplication from 1,013 raw rows) across nine conflicts. Table 1 reports the full per-conflict extraction statistics; all nine conflicts were carried forward to network analysis.

Table 1: Per-conflict extraction statistics. Mfr cov. = share of assets with at least one P176 link.

Conflict	Assets	Mfrs	Mfr cov.
Iraq	130	116	94.9%
Ukraine	89	75	89.7%
Kosovo 1999	80	93	92.6%
Afghanistan	77	59	92.3%
Gaza	60	30	75.3%
Lebanon 2006	57	28	75.7%
Syria	34	27	83.3%
Nagorno-Karabakh	7	9	83.3%
Iran–Iraq	5	8	92.3%
Total	**539**	**194**	—

4.2 Ownership Reconstruction

The iterative BFS traversal converged after five iterations. The final raw ownership table comprised 642 rows; after removing 44 noise rows, the clean table contained 598 rows covering 192 manufacturers and 145 unique ultimate owners. Of the 194 manufacturers, 174 (89.7%) were linked to at least one non-self owner under *all_links* mode and 166 (85.6%) under *controlish* mode. The 20–28 self-terminal manufacturers (≈10–14%) are principally state-owned entities without explicit parent relations in Wikidata.

Under *all_links* mode, the top ultimate owners by manufacturer count are: Russia (42), BlackRock (24), United States (23), The Vanguard Group (22), France (13), and State Street Corporation (12). The appearance of institutional investors reflects Wikidata's encoding of minority equity holdings rather than operational control, a distinction addressed in Section 5.

4.3 Network Analysis and Centrality

The bipartite Owner–Conflict graph comprised 74 owner nodes, 9 conflict nodes, and 216 weighted edges. The projected owner–owner network comprised 74 nodes and 1,932 edges. The full directed ownership graph G (no P361) contained 431 nodes and 520 edges. Table 2 lists the *systemic players*; entities present in five or more conflicts under *controlish* mode.

Table 2: Systemic players: ultimate owners present in ≥5 conflicts (*controlish* mode). Degree (w) = weighted degree; PR = PageRank (α=0.85).

Owner	Conflicts	Deg.(w)	PR
United States	8	57	0.0560
Israel	6	16	0.0077
French State	6	12	0.0048
Russia	5	20	0.0376
Italy	5	17	0.0046
India	5	5	0.0018
Germany	5	6	0.0031

Among conglomerate-level entities (see Appendix Table A2), Rheinmetall AG (Germany) achieves the highest betweenness centrality (0.00047), followed by Northrop Grumman/US (0.00046), Penta Investments Group/Jersey (0.00030), and Oboronprom/Russia (0.00027). These entities act as structural bridges in the ownership graph; manufacturers of multiple asset classes across different conflicts pass through them. The top PageRank scores confirm state-level dominance: United States (0.0560), Russia (0.0376), and Rostec/Russia (0.0171) rank first, second, and third respectively (see Appendix Table A3).

4.4 Ownership Concentration (HHI)

At country level (*all_links*), Syria (0.574) and Nagorno-Karabakh (0.500) show the highest concentration, with the top country controlling 75% and 67% of mapped manufacturers respectively. Lebanon 2006 (0.384), Ukraine (0.306), and Gaza (0.330) form an intermediate band, while Kosovo 1999 (0.144) and Iraq (0.160) are the most fragmented. At conglomerate level (*controlish*), all values drop substantially: Iraq falls to 0.035 and Kosovo to 0.037; confirming that state-level concentration metrics overstate operational centralisation. Nagorno-Karabakh (0.250) and Syria (0.149) retain the highest conglomerate-level concentration, though with only few manufacturers in the sample this convergence should be interpreted with caution and may reflect data sparsity rather than a genuine structural feature. Table 3 reports all values.

Table 3: HHI at country level (*ctry*, *all_links*) and conglomerate level (*cng*, *controlish*), sorted by HHI_{ctry}. HHI=1 = monopoly.

Conflict	HHI_{ctry}	HHI_{cng}
Syria	0.574	0.149
Nagorno-Kar.	0.500	0.250
Lebanon 2006	0.384	0.125
Gaza	0.330	0.111
Ukraine	0.306	0.077
Afghanistan	0.281	0.058
Iran–Iraq	0.160	0.143
Iraq	0.160	0.035
Kosovo 1999	0.144	0.037

4.5 Community Detection

The Louvain algorithm partitioned the 74 owner nodes into three communities. C0 (29 nodes, Russia-dominant, 7 systemic entities) groups Russian state-controlled manufacturers including Rostec, United Aircraft Corporation, and Oboronprom; alongside Rheinmetall AG (German) and the Government of India, reflecting a shared supply presence in Soviet-era and non-Western conflict theatres. C1 (33 nodes, US-dominant, 11 systemic entities) is the largest community by systemic presence, grouping Northrop Grumman, the Italian Ministry of State Participations, alongside NATO-aligned actors. C2 (12 nodes, Israel-dominant, 4 systemic entities) clusters the United States, France, and Israel; consistent with their overlapping supplier presence in Gaza and Lebanon 2006. The three-community structure closely mirrors known geopolitical alignments.

4.6 Conflict–Conflict Jaccard Similarity

The most similar pair is Gaza and Lebanon 2006 (J = 0.923), indicating near-identical ownership profiles, consistent with both conflicts being heavily supplied by the same Israeli and French industrial actors. The second most similar pair is Iraq and Kosovo 1999 (J = 0.811), reflecting the dominant NATO-aligned supply chain common to both. A second cluster groups Afghanistan, Iraq, and Kosovo, all with pairwise similarities between 0.47 and 0.59, consistent with their shared Western supplier base. At the other extreme, Nagorno-Karabakh shares zero owners with Iran–Iraq, Lebanon 2006, and Syria (J = 0.000), indicating entirely disjoint ownership sets.

5. Discussion

5.1 Interpretation of Findings

The principal finding is that the United States and Russia function as systemic industrial actors across most sampled conflicts, with PageRank scores an order of magnitude above any other single conglomerate. This is consistent with SIPRI data showing US and Russian firms consistently accounting for over 60% of global arms revenues [2]. The novelty lies in demonstrating that this dominance is structurally encoded in the ownership graph: both countries appear as terminal controllers of manufacturers supplying opposing or unrelated belligerents within the same conflict. Importantly, the distinction is not only one of breadth but of depth: the United States appears in 8 conflicts with a weighted degree of 57, nearly three times Russia's weighted degree of 20 across 5 conflicts. This indicates that US ownership does not merely span more wars, but that it controls substantially more manufacturers per conflict, a structural asymmetry that aggregate presence counts alone would not reveal.

The appearance of institutional investors (BlackRock, Vanguard, State Street) among the top five ultimate owners under *all_links* mode requires careful interpretation. Wikidata encodes minority equity stakes (typically sub-10% holdings) under the same P127/P749 predicates used for majority control. These are passive financial holdings, not operational control relationships. The *controlish* traversal mode partially mitigates this conflation, but does not eliminate it entirely, given the inconsistently applied semantics of user-contributed Wikidata triples. This constitutes the primary limitation of a Wikidata-only approach and points to a clear direction for future work: supplementing with authoritative ownership databases (OpenCorporates, LEI Registry) to unambiguously separate beneficial ownership from portfolio investment.

The dramatic HHI gap between country and conglomerate levels, most visible for Ukraine (0.306 vs. 0.077) and Iraq (0.160 vs. 0.035), reveals a structurally important asymmetry. State-level concentration metrics, while useful for geopolitical analysis, overstate the degree of centralised operational control. Conglomerate-level analysis is therefore essential for understanding who actually directs production. An interesting pattern emerges when HHI and Jaccard results are read together. Lebanon 2006 and Gaza display near-identical concentration profiles at both levels: country (0.384 vs. 0.330) and conglomerate (0.125 vs. 0.111); with gaps of comparable magnitude in both cases. Given that their Jaccard similarity is 0.923, this parallel structure is not coincidental.

A structural distinction emerges between state actors and conglomerate-level entities when betweenness centrality is considered: every sovereign state in the table (US, Russia, France, French State, Israel, Italy, India, Germany) has betweenness = 0.0000. Only conglomerate-level entities have non-zero betweenness (Northrop, Rheinmetall, General Dynamics, Oboronprom). States are terminal nodes that absorb ownership chains but do not bridge them. The bridging function in the graph belongs entirely to private or semi-private conglomerates. A further illustration of this dynamic is Rostec, which connects to more manufacturers per conflict than Russia does as a direct owner (weighted degree (21) exceeds Russia's), indicating that Russian state industrial influence is channeled primarily through Rostec as an operational intermediary rather than through direct state ownership. Russia's high PageRank (0.0376) reflects recursive influence through Rostec, not direct manufacturer control.

The Jaccard similarity matrix reveals a pattern that partially decouples from the community structure. Syria presents a structurally ambiguous profile: it shows moderate ownership similarity with Afghanistan ($J = 0.43$) and Iraq ($J = 0.26$), suggesting a partial overlap with the Western supply chain, yet shares zero owners with Nagorno-Karabakh despite both conflicts having significant Russian industrial involvement. This divergence indicates that Syrian ownership draws on a mixed supplier base that community detection, which aggregates co-occurrence across all conflicts, smooths over. It is a reminder that similarity at the conflict-pair level can reveal local supply-chain overlaps that cluster-level analysis obscures.

The Ward hierarchical clustering underlying the row and column ordering of the Jaccard similarity matrix reveals two macro-clusters: a Middle Eastern supplier bloc grouping Gaza and Lebanon 2006, and a broader Western-supply cluster grouping Iraq, Kosovo 1999, Afghanistan, Syria, and Ukraine. Iran–Iraq and Nagorno-Karabakh fall outside both clusters as structural outliers, sharing minimal ownership overlap with any other conflict in the sample. This two-bloc structure mirrors the community partition recovered by Louvain, providing independent corroboration from a different analytical method.

The community structure recovered by the Louvain algorithm is analytically significant precisely because it was derived from ownership relations alone, with no geopolitical input: the algorithm had no access to alliance memberships, conflict alignments, or bilateral trade data, yet partitioned owners into clusters that closely mirror known Cold War and post-Cold War industrial blocs. Two features of the clustering warrant closer interpretive attention. First, the presence of Rheinmetall AG and the Government of India within the Russia-dominant community C0 is not an artefact of noise: both entities share a supply presence in Soviet-era and non-Western conflict theatres, the same operational environment that defines C0's cohesion, without being organisationally linked to Russian state industry. This illustrates a limitation of community detection as an analytical tool: co-occurrence in conflicts drives cluster assignment, so actors with historically diversified portfolios may be placed in communities that do not reflect their primary political alignment. Second, the Israel-dominant community C2 groups the United States, France, and Israel despite the fact that US and French entities appear as systemic players across nearly

all conflicts. Their placement in C2 reflects the particular density of their co-occurrence in Gaza and Lebanon 2006, where overlapping supplier profiles are strongest. This structural correspondence suggests that corporate ownership hierarchies, even as encoded in an incomplete linked-data source like Wikidata, carry latent information about industrial-political alignment that is not explicitly represented anywhere in the graph.

5.2 Challenges and Limitations

The most significant challenge was Wikidata coverage heterogeneity. Asset density varies enormously across conflicts: Iraq yields 130 unique weapons versus 7 for Nagorno-Karabakh, reflecting the differential depth of Wikidata's curation effort rather than real-world differences in industrial complexity. This imbalance affects the reliability of HHI comparisons between data-rich and data-sparse conflicts and should be considered when interpreting concentration metrics for smaller or less-documented wars.

The BFS depth cap at five levels is a pragmatic choice that trades completeness for reproducibility under WDQS constraints. Some ownership chains, particularly for smaller or non-Western manufacturers, may not reach their true ultimate controller within five hops. The fallback proxy partially compensates but conflates country of headquarters with beneficial ownership.

A further structural limitation concerns the temporal validity of the ownership data. Wikidata encodes ownership relations as they exist at the time of extraction (March 2026), not as they existed during the conflicts under study, some of which span decades (Iran–Iraq War, 1980–1988; Iraq War, 2003–2011; Afghanistan, 2001–2021). Corporate hierarchies evolve through mergers, acquisitions, nationalisations, and divestitures; an ownership chain that is accurate today may not reflect the controlling structure at the time the relevant weapons were manufactured or deployed. This limits the causal interpretation of results: the study maps current ownership onto historical conflicts, and should therefore be understood as revealing structural industrial alignments rather than making claims about corporate intent or influence at the time of specific events.

5.3 Contribution and Future Work

This project makes three contributions: (i) a replicable SPARQL-to-graph pipeline for cross-conflict ownership mapping; (ii) the first empirical HHI-based comparison of defense ownership concentration across nine contemporaneous wars; and (iii) a demonstration that Louvain community structure on Wikidata-derived ownership graphs recovers geopolitically coherent clusters. The methodology is directly transferable to other domains where supply-chain ownership intersects geopolitical risk (critical minerals, semiconductors, energy infrastructure).

Three avenues for future work emerge naturally. First, integrating authoritative corporate ownership databases (LEI Registry, OpenCorporates) would disambiguate financial from operational control. Second, extending the pipeline to a larger, temporally stratified conflict set would enable longitudinal analysis of concentration changes during conflict escalation cycles. Third, enriching the RDF KG with SIPRI revenue data would allow weighted analysis reflecting actual production volume rather than manufacturer count.

6. Conclusions

This project constructed a multi-conflict ownership knowledge graph linking 835 conflict–asset–manufacturer triples across nine contemporary wars to 145 ultimate controlling entities using Wikidata as the sole linked-data source. Network analysis identified the United States (8 conflicts) and Russia (5 conflicts) as the only systemic actors in the sample. HHI analysis revealed a wide concentration spectrum from near-monopoly in Syria to moderate fragmentation in Iraq, and demonstrated that conglomerate-level analysis systematically reveals lower concentration than country-level analysis. Community detection recovered three ownership clusters closely mirroring known geopolitical alignments.

The study demonstrates the analytical power of knowledge graphs for tracing industrial influence in complex global systems and provides an empirical foundation for researchers, policymakers, and civil society organisations seeking to understand the corporate architecture behind contemporary warfare. The full pipeline, code, RDF graph, and visualisations are documented and reproducible from the accompanying project ZIP.

References

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7. AI Use Declaration

The author used AI-assisted tools solely for two purposes: elaborating and stress-testing analytical ideas through dialogue, and checking grammatical correctness of the English text. All research design, data extraction, pipeline implementation, analysis, and intellectual contributions are entirely the author's own.

Appendix — Supplementary Tables and Figures

owner	country	n_conflicts	degree_w	pagerank	betweenness	in_degree
United States	United States	8	57	0.0560	0.0000	32
French State	France	6	12	0.0048	0.0000	4
Israel	Israel	6	16	0.0077	0.0000	5
France	France	6	15	0.0148	0.0000	9
Northrop Grumman Corporation	United States	5	11	0.0035	0.0005	3
Ministry of Economy and Finance	Italy	5	17	0.0046	0.0001	3
Government of India	India	5	5	0.0018	0.0000	1
Ministry of State Participations	Italy	5	17	0.0037	0.0000	1
Rheinmetall AG	Germany	5	6	0.0031	0.0005	3
Russia	Russia	5	20	0.0376	0.0000	17
Finmeccanica	Italy	4	4	0.0014	0.0000	1
Walloon Region	Belgium	4	5	0.0029	0.0000	1
General Dynamics	United States	4	4	0.0025	0.0003	1
Lockheed Martin	United States	4	7	0.0034	0.0001	3
Germany	Germany	4	11	0.0129	0.0000	7
Government of Argentina	Argentina	4	4	0.0018	0.0000	1
General Motors	United States	4	12	0.0031	0.0001	3
Oboronprom	Russia	4	12	0.0034	0.0003	2
Rolls-Royce	United Kingdom	4	4	0.0022	0.0002	1
Rostec	Russia	4	21	0.0171	0.0002	11

Table A1: Full centrality table for owners in ≥4 conflicts (*controlish* mode): weighted degree, PageRank, betweenness, and in-degree.

owner	country	betweenness	n_conflicts
Rheinmetall AG	Germany	0.00047	5
Northrop Grumman Corporation	United States	0.00046	5
Penta Investments Group Limited	Jersey	0.00030	3
General Dynamics	United States	0.00027	4
Oboronprom	Russia	0.00027	4
Exor	Netherlands	0.00024	2
Boeing	United States	0.00020	4
Rolls-Royce	United Kingdom	0.00018	4
Rostec	Russia	0.00017	4
United Aircraft Corporation	Russia	0.00016	3

Table A2: Top-10 owners by betweenness centrality in the full ownership graph G.

owner	country	pagerank	n_conflicts
United States	United States	0.05598	8
Russia	Russia	0.03764	5
Rostec	Russia	0.01712	4
France	France	0.01483	6
Germany	Germany	0.01291	4
Japan	Japan	0.00984	2
United Aircraft Corporation	Russia	0.00925	3
Ukraine	Ukraine	0.00794	1
Israel	Israel	0.00770	6
Canada	Canada	0.00685	3

Table A3: Top-10 owners by PageRank ($\alpha = 0.85$) in the full ownership graph G.

HHI per conflict — Country level (all_links)

conflict	hhi	top1_share	top3_share	n_owners	n_manu
syria	0.5740	75.0%	85.7%	7	28
nagorno_karabakh	0.5000	66.7%	100.0%	3	6
lebanon_2006	0.3843	54.5%	90.9%	5	22
gaza	0.3299	50.0%	83.3%	6	24
ukraine	0.3058	52.9%	72.1%	16	68
afghanistan	0.2814	50.7%	65.7%	16	67
iran_iraq	0.1600	20.0%	60.0%	7	10
iraq	0.1596	35.3%	56.0%	25	116
kosovo_1999	0.1440	32.4%	52.9%	24	102

HHI per conflict — Conglomerate level (controlish)

conflict	hhi	top1_share	top3_share	n_owners	n_manu
nagorno_karabakh	0.2500	37.5%	75.0%	5	8
syria	0.1488	32.0%	52.0%	13	25
iran_iraq	0.1429	14.3%	42.9%	7	7
lebanon_2006	0.1247	21.1%	52.6%	12	19
gaza	0.1111	19.1%	47.6%	13	21
ukraine	0.0774	18.2%	40.3%	33	77
afghanistan	0.0579	16.7%	28.8%	31	66
kosovo_1999	0.0370	11.7%	21.4%	49	103
iraq	0.0353	11.1%	21.4%	53	126

Table A4: HHI per conflict at country level (*all_links*) and conglomerate level (*controlish*), with top-1 and top-3 ownership shares.

community	dominant	n_members	n_systemic	top owners
C0	Russia	29	7	Government of India, Rheinmetall AG, Russia
C1	United States	33	11	Northrop Grumman Corporation, Ministry of Economy and Finance, Ministry of State Participations
C2	Israel	12	4	United States, French State, Israel

Table A5: Louvain community summary: dominant country, number of members, systemic entities, and top owners per community.

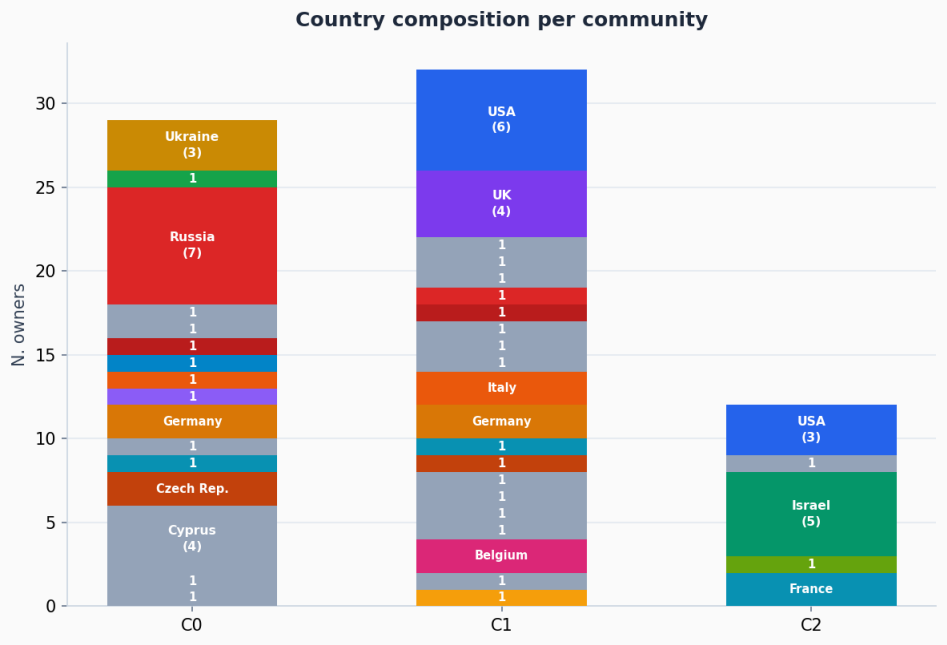


Figure A1: Country composition per Louvain community (C0 Russia-dominant, C1 US-dominant, C2 Israel-dominant).

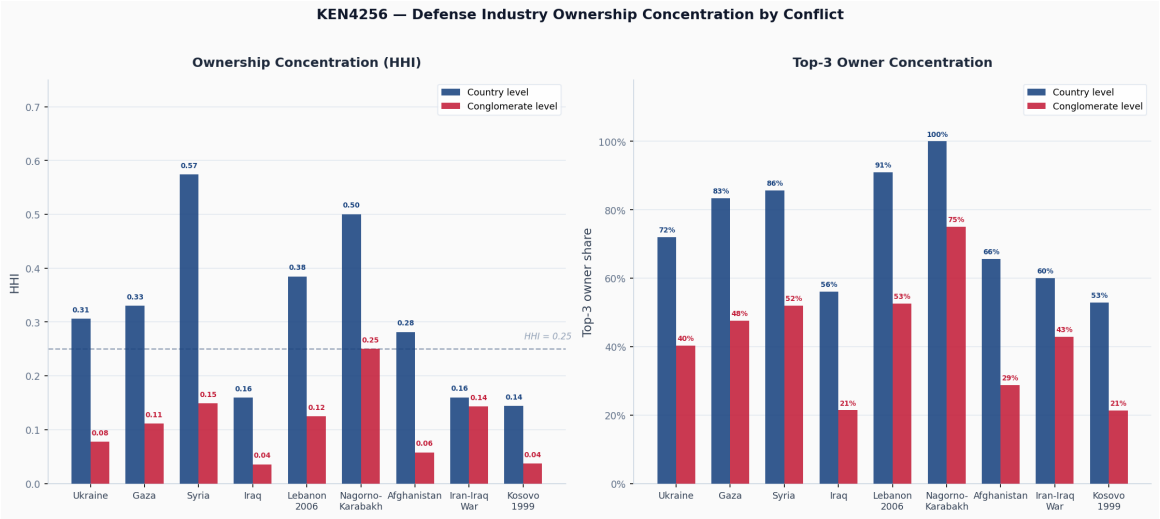


Figure A2: Ownership concentration (HHI) and top-3 owner share by conflict, at country and conglomerate level.

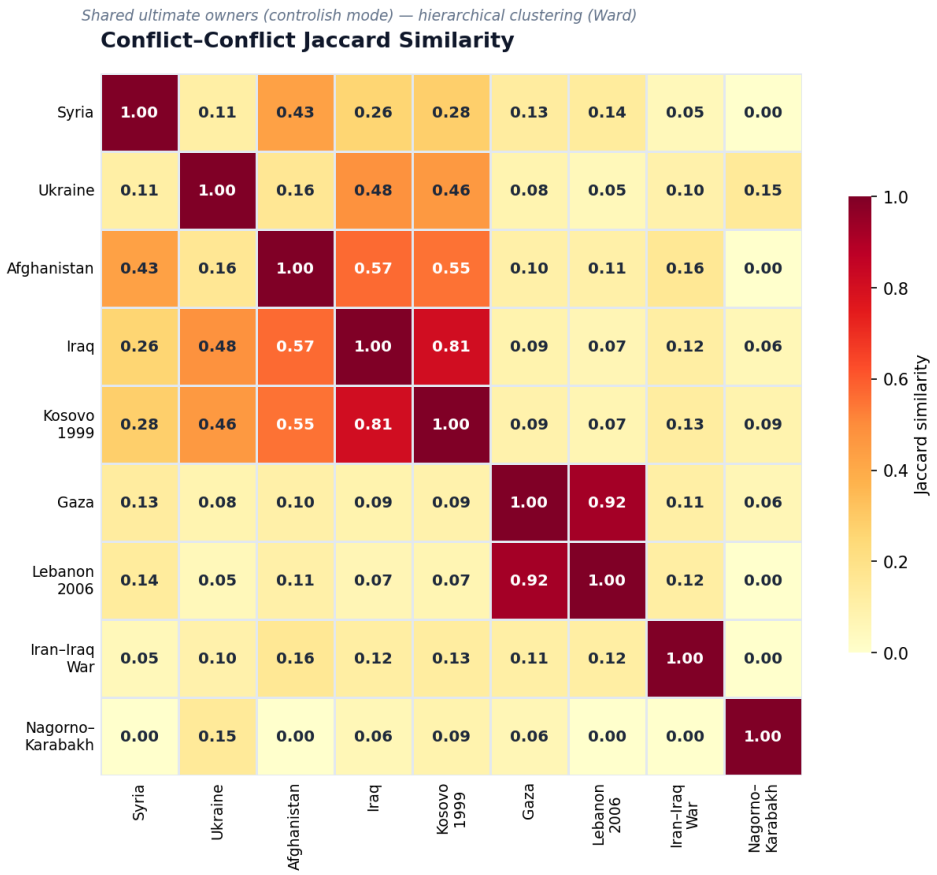


Figure A3: Conflict–conflict Jaccard similarity matrix (*controlish* mode), ordered by Ward hierarchical clustering.

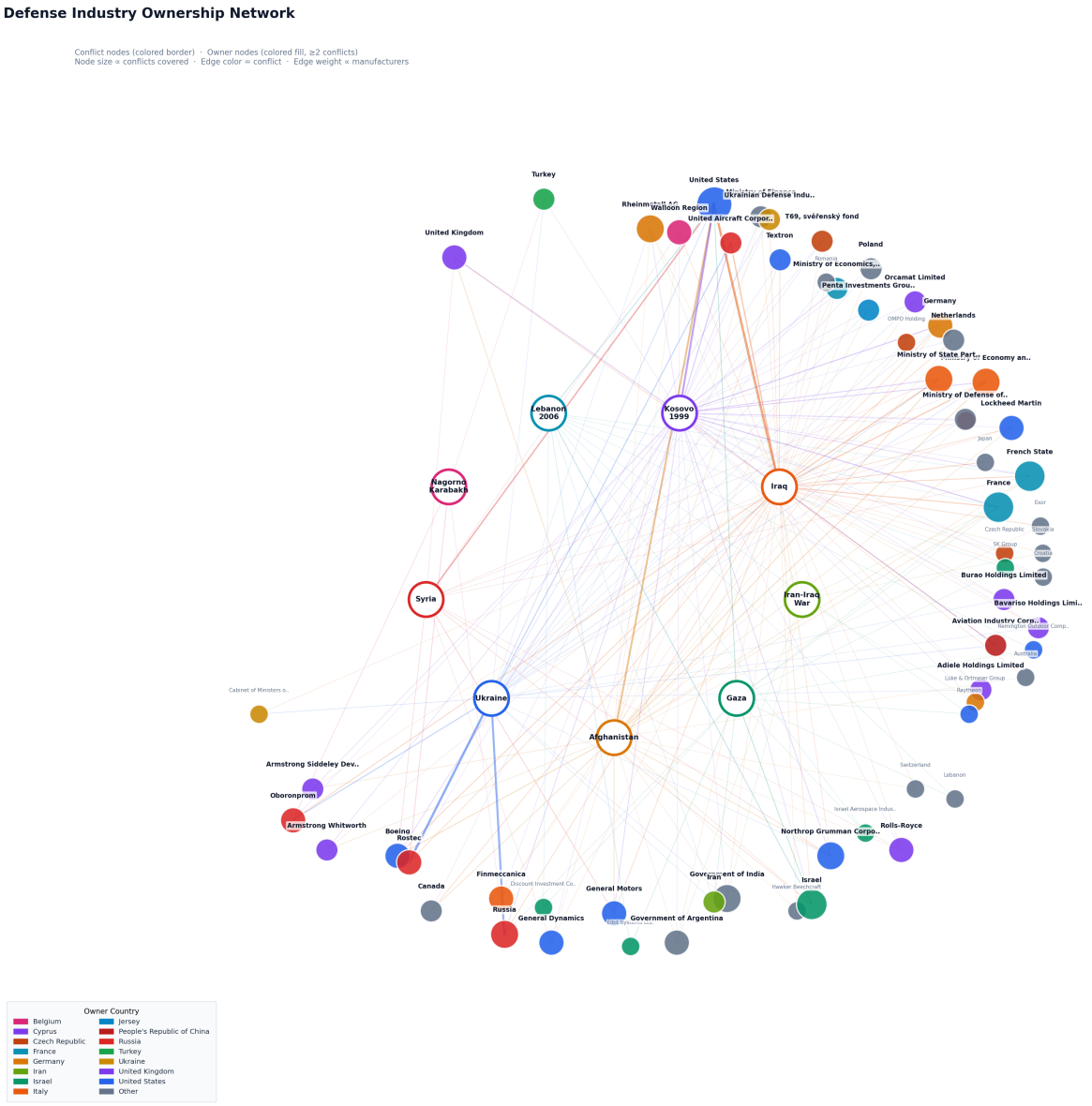


Figure A4: Defense industry ownership network. Conflict nodes have coloured borders; owner nodes (in $\geq$2 conflicts) are filled by country. Node size $\propto$ conflicts covered; edge weight $\propto$ manufacturers.